

What Can Three Seeds and Fifty Rollouts Resolve? A Preregistered Audit of Operator and Seed Variance on robomimic Multi-Human Benchmarks

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Abstract—Results on the robomimic Multi-Human (MH) imitation-learning benchmarks are conventionally reported as a mean over three training seeds, each evaluated with 50 rollouts. We preregistered the hypothesis that the variance caused by *which operators’ demonstrations are in the training set* exceeds the variance caused by training seed, and tested it with 138 BC-RNN training runs on Lift, Can and Square: ten seeds at fixed data, six leave-one-operator-out partitions, and six size-matched random partitions as a control, all evaluated at the final checkpoint on fixed initial states. The preregistered tests were not met on any task, and we report that negative result with its power: with six operators the procedure detects a true ratio of two only 18% of the time. What the data do establish concerns the reporting convention itself. With 50 rollouts on a shared set of initial states, checkpoints trained with different seeds cannot be told apart by overall success rate on any task; they differ mainly in which states they solve. On Square, two three-seed means must differ by 12.7 points to be distinguished. Re-evaluating the same 46 Square checkpoints with 500 rollouts, the seed effect becomes detectable, the 50-rollout success rates correlate only $r = 0.56$ with the 500-rollout ones, and an apparent proficiency-tier pattern (between-tier share 0.94) disappears (0.43). At 500 rollouts the operator-partition standard deviation is 0.054 against 0.014 for random partitions. That contrast nominally meets the preregistered threshold but does not survive Monte Carlo error, and we do not treat it as established. We give concrete reporting recommendations.

Index Terms—Imitation learning, benchmarking, reproducibility, variance components, learning from demonstration.

I. INTRODUCTION

THE robomimic study [1] established the most widely used protocol for evaluating offline imitation learning in manipulation: train on a fixed demonstration dataset, evaluate with 50 rollouts, and report success rate averaged over three seeds. Its Multi-Human (MH) datasets, 300 demonstrations per task from six operators of varied proficiency collected with RoboTurk [2], are a standard testbed, and later work reports on them with the same three training seeds and 50 evaluation episodes, for example Diffusion Policy and the baselines it compares against [3]. The three-seed standard deviation is the only uncertainty attached to such numbers.

A three-seed standard deviation measures one thing: how much the result moves when the optimizer’s random stream changes and everything else is held fixed. It says nothing about how much the result would move had the dataset

contained different demonstrations, or demonstrations from different people. For a benchmark built from six operators, that second source is not obviously smaller than the first. If it is larger, the conventional uncertainty understates the real one, and comparisons separated by a few points are not resolvable from the reported statistic.

We set out to test that claim directly and preregistered it before any training run. The claim, as preregistered, was not supported. This paper reports that outcome plainly, and then reports what the experiment did establish, which is a statement about the measurement rather than about operators:

- **A preregistered negative result with its power.** On no task does the operator-partition standard deviation exceed the seed standard deviation, nor does it exceed the matched random-partition control, under the preregistered tests (Section IV-B). With six operators the preregistered procedure has 18% power at a true ratio of two and 42% at three, so the negative result bounds little.
- **At 50 rollouts the seed effect is invisible.** On a shared set of 50 initial states, an exact paired test finds no difference in overall success rate between ten seeds on any task. The checkpoints do differ, but in which states they solve; that difference only averages out over many states. At 500 states the seed effect on Square is detectable (Section IV-A).
- **Fifty rollouts do not characterise a checkpoint.** Re-evaluating the same 46 Square checkpoints with 500 rollouts, the 50-rollout numbers explain 31% of the variance of the 500-rollout numbers, and a tier pattern that looked strong at 50 rollouts is absent at 500 (Section IV-C).
- **A resolvability table.** We give the smallest difference between two three-seed means that the convention can resolve on each task, for comparisons on shared and on independent initial states (Section IV-E), and recommendations that follow (Section V).

We cite robomimic as the origin of a convention that the field, including this author, has used; nothing here is a criticism of that study, which also released the per-operator labels that make this audit possible.

II. RELATED WORK

Seeds and statistical practice. That deep reinforcement learning results vary strongly with seed, and that small seed counts make comparisons unreliable, is well documented [4],

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Code, preregistration, partitions, initial states, per-run and per-rollout results: <https://github.com/imjbassi/operator-variance-audit>, archived at <https://doi.org/10.5281/zenodo.23109850>.

[5], [6]. Bouthillier *et al.* [7] showed for supervised benchmarks that variance from data sampling, parameter initialisation and hyperparameter choice all markedly affect comparisons, and that estimators which account for more of these sources are more reliable; our Arm C is the imitation-learning analogue of their data-resampling source. For robot learning, Kress-Gazit *et al.* [8] recommend reporting evaluation conditions and statistical uncertainty in detail. We add a measurement, on a specific and heavily used benchmark, of how the conventional statistic decomposes.

Demonstrator heterogeneity. The robomimic study analysed MH data by proficiency tier, pooling the two operators in each tier, to ask how algorithms cope with suboptimal data [1]. Individual operators were not a unit of analysis there. Data quality and the compatibility of demonstrations from different people are known to affect imitation learning [9], [10]. Our question is narrower and statistical: treating the operator as the unit of replication, how much does benchmark performance move when one operator is removed, relative to seed and to removing the same number of random demonstrations?

Preregistration. Fixing hypotheses and analyses before data collection separates confirmatory from exploratory findings [11]. We follow that practice, including a dated amendment log, and keep the distinction explicit throughout.

III. PREREGISTERED DESIGN

The preregistration, with predictions, falsification criteria and the analysis plan, was committed to the public repository before any training run and is reproduced there unchanged, followed by four dated amendments.

A. Data and operator identity

We use the robomimic v1.5 MH low-dimensional datasets for Lift, Can and Square (robosuite [12] 1.5.1). Each has 300 demonstrations, 50 from each of six operators. Operator identity is read directly from the filter keys shipped in the files (`better_operator_1`, ..., `worse_operator_2`); we verified that the six keys are disjoint, each holds 50 contiguous demonstrations, and their union is the full dataset. No inference of identity is involved. The operator index is defined within a tier, and whether a given label denotes the same person across tasks is not documented, so we make no cross-task statement about individuals.

B. Policy, training and evaluation

All runs train BC-RNN with the robomimic low-dimensional hyperparameters: a two-layer LSTM of width 400 with a five-mode Gaussian-mixture head, sequence length 10, Adam with learning rate 10^{-4} , batch size 100, and 2000 epochs of 100 gradient steps (robomimic v0.5.0). We evaluate only the *final* checkpoint. The original study reports the maximum success rate over periodic evaluations during training and notes that other selection rules give lower numbers [1]; selecting on the evaluation rollouts is selection on the test set, so no selection is performed here. Our final-checkpoint means on the full datasets are 0.988, 0.946 and 0.600. For reference,

the maximum-over-training figures reported in [1, Table 1] for BC-RNN on the original MH low-dimensional datasets are 100.0, 100.0 and 78.0; the two sets of numbers differ in checkpoint rule and dataset version and are not directly comparable.

Each checkpoint is evaluated with 50 rollouts of horizon 500 on 50 initial conditions that are generated once per task from seeded resets, stored, and reused for every checkpoint. Because robosuite regenerates the scene model at each reset and Lift randomises the cube size, both the simulator state and the model description are stored and restored. The rollout random stream is also fixed per rollout index. Evaluation is repeatable: re-running a checkpoint on the stored conditions reproduced every rollout outcome and step count in our check.

C. Three arms

Arm A (seed). All 300 demonstrations, ten seeds: 10 runs per task.

Arm B (operator composition). Six leave-one-operator-out partitions of 250 demonstrations, three seeds each: 18 runs per task. All partitions have the same size with no subsampling.

Arm C (matched random control). Six partitions of 250 demonstrations drawn uniformly at random from the 300, ignoring operators, three seeds each: 18 runs per task.

Arm C is load-bearing. Two Arm B partitions differ in which operator is absent, but also simply in which demonstrations are present; any two of them share 200 of their 250 demonstrations. Arm C partitions have the same size and a comparable overlap, and isolate the second effect. If the spread across Arm B partitions does not exceed the spread across Arm C partitions, an operator effect is not separable from demonstration resampling at this sample size. Partition memberships were generated once, committed, and never changed. In total there are 138 training runs and 6,900 evaluation rollouts.

D. Statistics

The unit of replication is the operator (equivalently, the partition), never the rollout or the demonstration. The per-run statistic is the success rate y of the final checkpoint.

Seed spread. For Arm A we report the sample standard deviation over the ten seeds, s_A , with a seed-bootstrap interval. s_A is the spread of success rates *measured on the shared stored states*. Two floors are relevant to it. The binomial floor $\sqrt{\bar{p}(1-\bar{p})/n}$ is the sampling error of one checkpoint's success rate against a fresh draw of n initial states. On shared states, however, outcomes are correlated across checkpoints, since easy states are solved by most of them. The appropriate null for "no difference between seeds" is then the per-state floor $\sqrt{q_i(1-q_i)/n}$, where q_i is the success rate on state i across checkpoints, and the exact test is Cochran's Q on the checkpoints-by-states success matrix. The per-state floor and Cochran's Q were added after review (Amendment 4) and are not preregistered.

Variance components. For Arms B and C separately, the 6×3 table of partitions p by seeds s is modelled as

$$y_{ps} = \mu + a_p + b_s + e_{ps}, \quad (1)$$

with independent zero-mean effects of variance σ_p^2 , σ_s^2 and σ_e^2 . We estimate the components by the method of moments, $\hat{\sigma}_p^2 = (\text{MS}_p - \text{MS}_e)/3$ and $\hat{\sigma}_s^2 = (\text{MS}_s - \text{MS}_e)/6$, truncated at zero [13], and, as a second estimator, by REML [14] in a crossed mixed model [15]. The residual absorbs rollout noise and any partition-by-seed interaction. An estimate of exactly zero is a boundary estimate and is marked as such; it is not evidence that the variance is zero.

Headline ratio. $R = \hat{\sigma}_p^B / s_A$, with a 95% percentile interval and a one-sided p -value for $R \leq 1$ from a joint bootstrap [16] that resamples the six partitions and the ten Arm A seeds with replacement (2000 resamples).

Control contrast. $D = (\hat{\sigma}_p^B)^2 - (\hat{\sigma}_p^C)^2$, with an interval and a one-sided p -value for $D \leq 0$ from a bootstrap over the six partitions of each arm. $D > 0$ means the operator partitions vary more than random partitions of the same size.

Decision rule and multiple comparisons. The primary family is six contrasts, $R > 1$ and $D > 0$ on three tasks, corrected by Holm's procedure [17]. A contrast is supported if its Holm-adjusted one-sided p -value is below 0.05. Intervals are shown unadjusted, for description. The preregistration stated an interval criterion for each contrast and a Holm correction for the family without saying which governs; Amendment 4 fixes the rule as stated here. Raw and corrected p -values are reported throughout.

Secondary claim. Tier labels were predicted not to capture the operator effect. We decompose the between-partition sum of squares of the Arm B partition means into between-tier and within-tier parts and compare the between-tier share to its distribution under relabelling.

E. Amendments

Amendment 1 was made before any data existed, after tests of the analysis code on synthetic tables. It records that the tier permutation test has only 15 distinct outcomes, so its p -value cannot fall below $1/15$ and the secondary claim must be judged by effect size; and that power would be reported by simulating the actual headline procedure, which synthetic runs had shown to be weak with six groups.

Amendment 2 was written after all Stage 1 results were seen and before any further rollout. Stage 1 showed that rollout noise was as large as the quantities being compared. Amendment 2 specifies a follow-up, *Stage 2*: the 46 existing Square final checkpoints are re-evaluated with 500 rollouts each on 500 new fixed initial conditions, and the same analysis is applied unchanged, with a two-contrast Holm family. Stage 2 reuses the Stage 1 checkpoints, was designed after seeing Stage 1, and is therefore a follow-up rather than independent confirmation. It also declares three analyses exploratory (Section IV-F).

Amendment 3 records the Stage 2 outcome and two reporting corrections described in Sections IV-C and IV-D.

Amendment 4 was made after external review of a draft, with all results known. It fixes the decision rule above, replaces the binomial floor by the per-state floor and Cochran's Q as the null for a seed effect on shared states, adds resolvability thresholds for comparisons on independent initial states, and records that the Stage 1 tier pattern was a two-operator pattern. These analyses are post hoc and labelled as such.

IV. RESULTS

A. The seed effect at 50 rollouts

Fig. 1 shows every run, and Table I the estimates. At fixed data, the ten-seed standard deviation is 0.017 on Lift, 0.023 on Can and 0.079 on Square, where the ten seeds range over 22 points (0.50 to 0.72). Table III compares these spreads with the floors of Section III-D.

No task shows a detectable seed effect at 50 rollouts. Cochran's Q on the ten-checkpoint by 50-state success matrix gives $p = 0.213$ on Lift, 0.855 on Can and 0.133 on Square. Against the binomial floor the picture is the same: on Can the spread is below it, and on Lift and Square it is above it by amounts a χ_9^2 test does not distinguish from zero ($p = 0.290$ and 0.223). The per-state floor is barely smaller than the binomial one here (0.064 against 0.069 on Square), so the two nulls agree.

The checkpoints are not identical, however. On Square 46 of the 50 states are solved by some seeds and not others. Seeds produce checkpoints that differ far more in *which* states they solve than in how many, and that difference only averages out over many states. A three-seed standard deviation computed from 50 rollouts therefore cannot see the seed effect on any of these tasks. We return to this at 500 states in Section IV-C.

B. Preregistered primary tests (Stage 1)

The primary claim is not supported on any task. The headline ratio is $R = 1.08$ on Square with interval $[0.00, 1.84]$, 0.66 on Lift and 0.18 on Can; all Holm-adjusted p -values are 1.000. The control contrast is positive on Square, with partition standard deviations of 0.086 in Arm B and 0.033 in Arm C, but it is not supported ($p = 0.0810$, Holm 0.4860, interval including zero). On Can the random partitions vary more than the leave-one-operator-out partitions (0.024 against 0.004). On Lift several components are boundary estimates, as expected for a task at ceiling. The two estimators agree on every component.

This negative result is weak evidence about effect size. Simulating the preregistered procedure at the fitted noise levels (Fig. 2), the probability that the interval excludes one is 18% when the true ratio is two, 42% when it is three and 66% when it is four, on Square, and no better on the other tasks. Two features of the design cause this. Six operators give a partition variance with five degrees of freedom, and a bootstrap over six clusters is known to be unreliable [18]. In addition, the preregistered denominator s_A contains the rollout noise floor, whereas the numerator is a residual-corrected component, so the ratio is conservative by construction. The simulated false-positive rate at a true ratio of one is at most 0.5% on any task and stage, against a nominal 2.5%. We noted the noise floor when Arm A finished, recorded its consequence for the ratio in dated notes before any Arm C result existed, and did not alter the test.

C. Stage 2: the same checkpoints at 500 rollouts

Fig. 3 shows the 46 Square checkpoints evaluated at 50 and at 500 rollouts. Three observations follow.

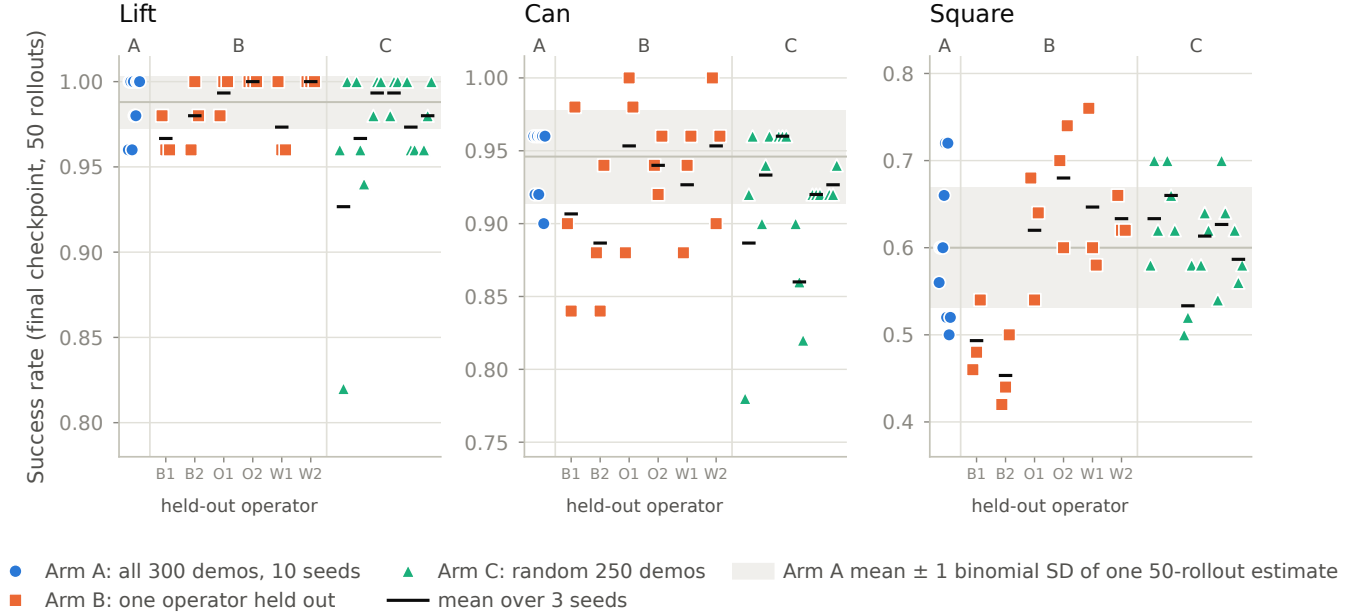


Fig. 1. Stage 1. Final-checkpoint success rate of every run (50 rollouts each). A: ten seeds on all 300 demonstrations. B: six leave-one-operator-out partitions, three seeds each, labelled by the held-out operator (B/O/W: better/okay/worse tier). C: six random 250-demonstration partitions, three seeds each. Bars are three-seed means. The grey band is the Arm A mean $\pm$ one binomial standard deviation of a single 50-rollout estimate. Note the different vertical scales.

TABLE I

VARIANCE COMPONENTS (METHOD OF MOMENTS; REML AGREES TO WITHIN 0.003 ON EVERY ENTRY) AND THE PREREGISTERED CONTRASTS. ARM A SD IS SHOWN WITH ITS 95% SEED-BOOTSTRAP INTERVAL. “FLOOR” IS THE BINOMIAL SD OF ONE ESTIMATE AT THE ARM A MEAN. $R = \hat{\sigma}_p^B / s_A$; $D = (\hat{\sigma}_p^B)^2 - (\hat{\sigma}_p^C)^2$. INTERVALS ARE 95% BOOTSTRAP INTERVALS FROM THE PREREGISTERED 2000 RESAMPLES. FOR D , p -VALUES ARE GIVEN AS PREREGISTERED RESAMPLING / EXACT ENUMERATION. $\dagger$ BOUNDARY ESTIMATE (TRUNCATED AT ZERO), NOT EVIDENCE OF ZERO VARIANCE.

Task	Arm A (10 seeds)			Arm B (operator held out)			Arm C (random 250)			R (Sec. III-D)		$D \times 10^3$ (Sec. III-D)		
	mean	SD	floor	$\hat{\sigma}_p$	$\hat{\sigma}_s$	$\hat{\sigma}_e$	$\hat{\sigma}_p$	$\hat{\sigma}_s$	$\hat{\sigma}_e$	est. [95%]	p_{Holm}	est. [95%]	p	p_{Holm}
<i>Stage 1 (preregistered primary): 50 rollouts per checkpoint</i>														
Lift	0.988	0.017 [0.006, 0.020]	0.015	0.011	0.000 $\dagger$	0.015	0.000 $\dagger$	0.000 $\dagger$	0.043	0.66 [0.00, 1.35]	1.000	0.12 [-0.15, 0.25]	0.1880 / 0.1937	0.9400 / 0.9684
Can	0.946	0.023 [0.000, 0.029]	0.032	0.004	0.025	0.046	0.024	0.000 $\dagger$	0.045	0.18 [0.00, 1.64]	1.000	-0.58 [-1.71, 0.44]	0.8010 / 0.7933	1.0000 / 1.0000
Square	0.600	0.079 [0.044, 0.096]	0.069	0.086	0.027	0.057	0.033	0.007	0.050	1.08 [0.00, 1.84]	1.000	6.22 [-1.35, 10.74]	0.0810 / 0.0844	0.4860 / 0.5062
<i>Stage 2 (follow-up, Amendment 2): same checkpoints, 500 rollouts</i>														
Square	0.639	0.031 [0.017, 0.039]	0.021	0.054	0.000 $\dagger$	0.045	0.014	0.010	0.045	1.71 [0.51, 2.96]	0.115	2.69 [0.02, 5.12]	0.0245 / 0.0251	0.0490 / 0.0501

A 50-rollout success rate is a poor measurement of a checkpoint. Across the 46 checkpoints the correlation between the two evaluations is $r = 0.56$: the 50-rollout number explains 31% of the variance of the 500-rollout number. The differences have standard deviation 0.068, matching the binomial expectation of 0.071, and reach 16 points for a single checkpoint. The 50-state set was also 3.3 points harder on average than the 500-state set, a reminder that results on a small fixed set of initial conditions are conditional on that set.

The seed effect becomes visible. At 500 rollouts the ten-seed standard deviation on Square is 0.031 [0.017, 0.039], against 0.079 at 50 rollouts, with a binomial floor of 0.021 and a per-state floor of 0.020. Cochran’s Q is now 22.0 on 9 degrees of freedom ($p = 0.009$): seeds produce checkpoints with genuinely different success rates, with a standard deviation of about 0.024 once the per-state floor is removed. Fifty states could not resolve this.

The operator contrast nominally meets the preregistered threshold, and we do not treat it as established. The partition standard deviation is 0.054 in Arm B and 0.014 in Arm C,

with equal residuals (0.045 and 0.045). The headline ratio is $R = 1.71$ [0.51, 2.96], which is not supported ($p = 0.115$). For the control contrast, the preregistered 2000-resample bootstrap gives $D \times 10^3 = 2.69$ [0.02, 5.12] with $p = 0.0245$ and Holm-adjusted 0.0490. *As preregistered, this computation meets the threshold*, under the decision rule of Section III-D and also under an unadjusted interval criterion. It does not do so robustly. With 2000 resamples the Monte Carlo standard error of that p -value is 0.0035, and of the Holm-adjusted value 0.0069, so the margin of 0.001 is well inside resampling noise. After seeing this result we enumerated the bootstrap exactly; there are only 462 distinct resamples of six partitions, so the full distribution can be computed with multinomial weights. The exact computation gives $p = 0.0251$, Holm-adjusted 0.0501, and an interval of $[-0.00, 5.23]$. Treating the contrast as not established is therefore a conservative choice made after the fact, and we state it as such. The point estimates in both stages have the operator partitions varying more than the random ones, and the data are consistent with an operator-composition effect several times larger than the resampling

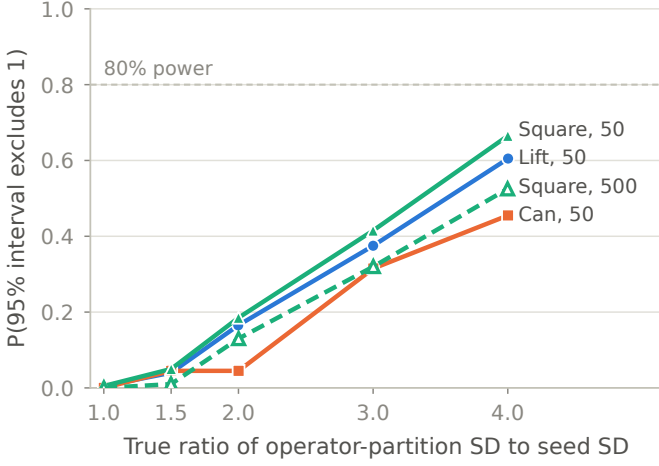


Fig. 2. Power of the preregistered headline procedure (probability that the 95% interval for R excludes one) as a function of the true ratio, simulated at each task’s fitted seed and residual standard deviations with six partitions, three seeds and ten Arm A seeds. Solid: 50 rollouts. Dashed: Square at 500 rollouts. Each point is 200 simulations, so its Monte Carlo standard error is up to 3.5 points.

effect, and equally with none that six operators can establish.

More rollouts lower power at a fixed ratio. At a true ratio of two the simulated power on Square is 18% at 50 rollouts and 13% at 500 (Fig. 2). The ratio is defined against s_A , which shrank by more than half, so a ratio of two now means a partition standard deviation of only 0.063, 1.40 times the Arm B residual, where at 50 rollouts it was 2.78 times. The residual did not shrink in proportion because it contains partition-by-seed interaction as well as rollout noise. Part of the difference is also simulation error, about 2.7 points per estimate.

D. Tier labels

At 50 rollouts the Square partition means looked like a clear tier effect: holding out either “better” operator gave 0.493 and 0.453, against 0.68 at most for the others, a between-tier share of 0.94 against a null mean of 0.40 (Table II). Even then it was not a tier pattern. The exact permutation p of 0.200 arises because exactly 3 of the 15 ways of pairing the six partitions reach the observed share, and those are precisely the pairings that keep the two “better” operators together; the actual tier pairing ranks third among them. Any grouping of the other four operators fits about as well, so the pattern was carried by two operators. After Stage 1 we recorded, in Amendment 2, that the secondary prediction appeared to be contradicted. At 500 rollouts the pattern is gone. The between-tier share is 0.43, 7 of the 15 pairings reach it, and the mean absolute difference between partitions in the same tier (0.075) is as large as between tiers (0.071). The partition whose mean moved most, with the first “better” operator held out, went from 0.493 to 0.624. Amendment 3 withdraws the earlier statement. We make no tier-level claim in either direction; what the comparison shows is that a pattern across six partition means, each a three-seed, 50-rollout average, was strong enough to be written down and did not survive ten times the rollouts.

TABLE II
SQUARE, ARM B: THREE-SEED MEAN SUCCESS RATE BY HELD-OUT OPERATOR AT 50 AND 500 ROLLOUTS, AND THE TIER EFFECT SIZES. THE EXACT PERMUTATION p FOR THE BETWEEN-TIER SHARE IS 0.200 AT 50 ROLLOUTS AND 0.467 AT 500; ITS MINIMUM ATTAINABLE VALUE IS 0.067.

Held-out operator	50 rollouts		500 rollouts	
	mean	seed range	mean	seed range
Better 1	0.493	0.46–0.54	0.624	0.62–0.63
Better 2	0.453	0.42–0.50	0.539	0.45–0.59
Okay 1	0.620	0.54–0.68	0.607	0.59–0.63
Okay 2	0.680	0.60–0.74	0.716	0.67–0.77
Worse 1	0.647	0.58–0.76	0.667	0.62–0.71
Worse 2	0.633	0.62–0.66	0.635	0.63–0.64
Between-tier share	0.94		0.43	
Null mean share	0.40		0.40	
Within-tier $ \Delta $	0.038		0.075	
Between-tier $ \Delta $	0.124		0.071	

E. What the convention can resolve

Arm A gives a direct estimate of the run-to-run standard deviation s of a reported number at fixed data. Two methods each reported as the mean of k seeds differ detectably, at a two-sided 95% level, only if their true difference exceeds roughly

$$\Delta_{\min} \approx 1.96 s \sqrt{2/k}, \quad s^2 \approx \sigma_c^2 + \frac{v}{n}, \quad (2)$$

where σ_c^2 is the variance of checkpoints’ true success rates across seeds and n is the number of rollouts. The noise term depends on how the two methods are evaluated. If both use the same stored initial states (paired), it is the per-state term $v = q_i(1 - q_i)$, and s is simply the seed spread s_A measured on shared states. If each evaluation draws its own initial states (unpaired), as with unseeded environment resets, it is $v = p(1 - p)$, which is never smaller. Table III gives both for $k = 3$. On Square at 50 rollouts the paired threshold is 12.7 points with a normal quantile, and 18.0 with the t quantile appropriate when s is itself estimated from the two three-seed samples; the unpaired threshold is 13.4. At 500 rollouts the paired threshold is 5.0 points. The paired and unpaired values differ most on Can, 3.7 against 5.1 points, where many states are solved by every checkpoint. These thresholds account for seed and rollout noise only. They do not include the dependence on which demonstrations are in the dataset, for which Arm C gives a partition standard deviation of 0.014 on Square at 500 rollouts, nor any operator effect.

F. Exploratory analyses

The following were declared exploratory in Amendment 2 and are not tests of the preregistered claims. (i) Replacing the denominator of R by the Arm A spread with the binomial floor removed gives 2.20 on Square at 50 rollouts and 2.36 at 500, but the correction is unstable: in 32% and 10% of resamples respectively the Arm A spread is at or below the floor and the ratio is undefined, and on Can it is undefined at the point estimate. (ii) The exact F -test for a partition component gives $p = 0.003$ in Arm B against 0.121 in Arm C on Square at

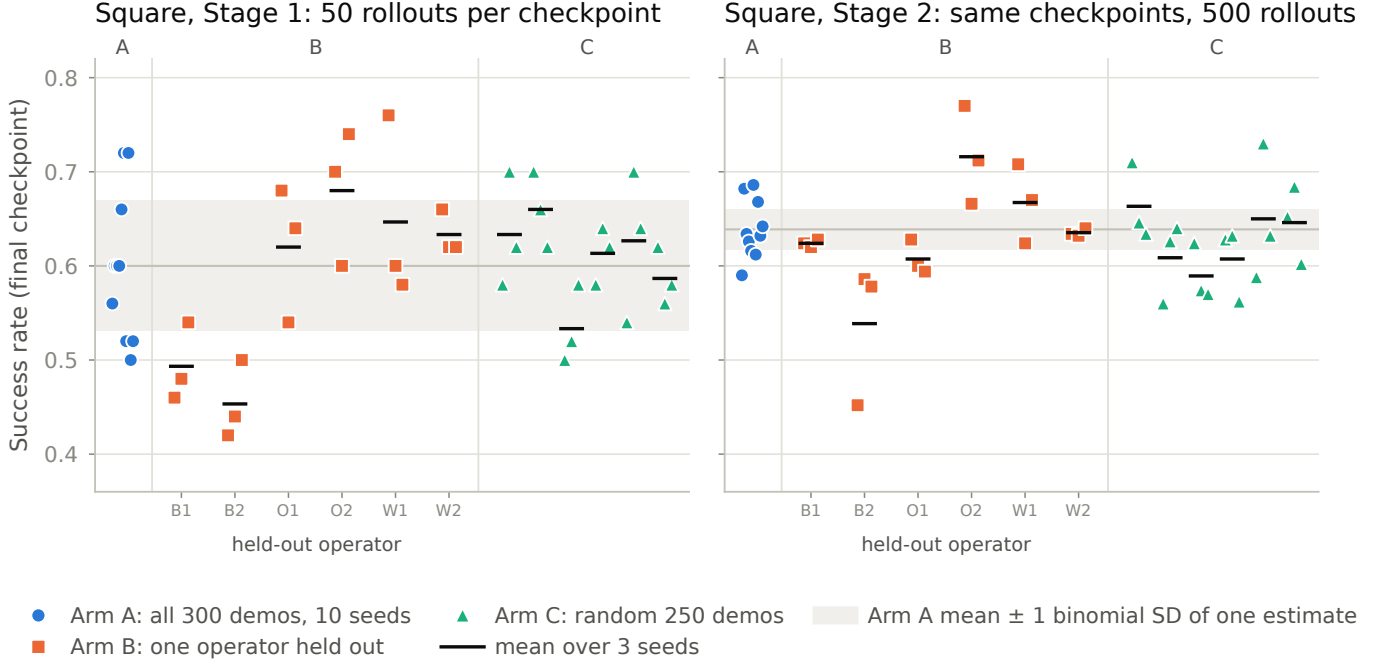


Fig. 3. Square: the same 46 final checkpoints evaluated with 50 rollouts (Stage 1, left) and with 500 rollouts on 500 new fixed initial conditions (Stage 2, right). Layout as in Fig. 1. The apparent separation of the two “better”-operator partitions at 50 rollouts is not present at 500.

TABLE III

SEED EFFECT AND RESOLVABILITY FROM ARM A (n INITIAL STATES, TEN SEEDS). FLOORS ARE THE BINOMIAL AND PER-STATE NOISE SDs OF SECTION III-D; COCHRAN’S p TESTS FOR ANY DIFFERENCE IN SUCCESS RATE BETWEEN THE TEN SEEDS ON SHARED STATES. THE LAST TWO COLUMNS GIVE THE SMALLEST DIFFERENCE BETWEEN TWO THREE-SEED MEANS DISTINGUISHABLE AT 95% (EQ. 2) WHEN BOTH METHODS USE THE SAME STORED STATES (PAIRED) OR INDEPENDENT DRAWS (UNPAIRED); IN PARENTHESES WITH THE t QUANTILE FOR 4 DEGREES OF FREEDOM. THE FLOORS, COCHRAN’S TEST AND THE UNPAIRED COLUMN ARE POST HOC (AMENDMENT 4).

Task	n	seed SD	floor		Cochran p	min. resolvable diff. (points)	
			binomial	per-state		paired	unpaired
Lift	50	0.017	0.015	0.015	0.213	2.7 (3.8)	2.8 (4.0)
Can	50	0.023	0.032	0.032	0.855	3.7 (5.3)	5.1 (7.2)
Square	50	0.079	0.069	0.064	0.133	12.7 (18.0)	13.4 (18.9)
Square	500	0.031	0.021	0.020	0.009	5.0 (7.1)	5.2 (7.3)

50 rollouts, and 0.012 against 0.349 at 500; on Lift and Can no test is below 0.05. (iii) The difference between partitions that hold out a “better” operator and the rest is 17.2 points at 50 rollouts and 7.5 at 500 on Square. These are consistent with a partition effect in Arm B on Square, and none of them changes the status of the preregistered contrasts.

V. DISCUSSION AND RECOMMENDATIONS

The hypothesis we registered was about operators. The clearest finding is about rollouts. With 50 rollouts, the evaluation cannot see differences between seeds on any of the three tasks, so a three-seed standard deviation reports the measurement more than the method. On Square the point estimate is also a weak proxy for the same checkpoint’s success rate under a larger evaluation. The published statement that motivated this study, that comparisons separated by a

few points are not resolvable from the reported statistic, is supported on Square by Table III, for a reason more elementary than the one we proposed.

For the operator question, both stages give the same ordering of point estimates on Square, with operator partitions varying more than size-matched random partitions, and neither stage establishes it. We regard this as an open question that six operators cannot settle at any rollout count: at 500 rollouts the preregistered procedure still has only 13% power at a true ratio of two. A dataset with more operators is required.

We recommend the following for results on these and similar benchmarks.

- 1) **Report the rollout noise next to the seed standard deviation, and test the seed effect on shared states.** A seed standard deviation near $\sqrt{p(1-p)/n}$ does not indicate a stable method; it indicates that the evaluation may not see the seed effect. When all checkpoints are evaluated on the same stored states, Cochran’s Q on the per-rollout outcomes is a simple exact test of whether they differ at all.
- 2) **Evaluate with enough rollouts for the task’s success band.** Near 60% success, 50 rollouts give a standard error of 7 points per checkpoint; 500 give 2. Rollouts in simulation are cheap relative to training, and Eq. (2) states what a given budget can resolve.
- 3) **Fix, store and publish the initial conditions**, including the scene model where the simulator randomises it, and report results on a set large enough that conclusions are not conditional on it.
- 4) **Evaluate a checkpoint chosen without the evaluation rollouts.** We used the final one.
- 5) **State the composition of the training set** when subsets

of a multi-operator dataset are used, and do not read structure, such as a tier effect, from a handful of low-rollout partition means.

- 6) **Release per-rollout outcomes.** They make every analysis above reproducible by others.

VI. LIMITATIONS

Six operators per task is the binding limitation; it sets the power of every operator-level test here. The study uses one architecture, low-dimensional observations, one training length, simulation only, and one benchmark family. Arms B and C have three seeds per partition. Partitions within an arm overlap heavily, so the partition effects are not independent draws from a population, and the random-effects model is an approximation; Arm C shares that structure, which is why the contrast with it, rather than the absolute component, is the meaningful quantity. Stage 2 was designed after Stage 1 results were known and reuses its checkpoints. The exact bootstrap enumeration was added after the Stage 2 contrast was seen to be at the boundary; it changes no statistic, but it is why two p -values are reported, and the preregistered computation, taken alone, would have supported that contrast. The per-state floor, Cochran’s test, the unpaired thresholds and the explicit decision rule were added after external review (Amendment 4); they change no preregistered decision at Stage 1. Operator labels are those released with the dataset, and no operator-level covariates exist. Success rates on a fixed set of initial conditions are conditional on that set.

VII. CONCLUSION

We preregistered the claim that operator composition contributes more variance than training seed on the robomimic Multi-Human benchmarks, ran 138 training runs with a matched random control, and did not confirm it. The experiment was underpowered for that claim by the number of operators, not by compute. It was well powered for a different conclusion: with 50 rollouts the conventional three-seed statistic cannot detect differences between seeds on any of the three tasks, cannot resolve differences between methods smaller than about 12.7 points on Square, and was sufficient to produce a tier pattern that a tenfold larger evaluation did not reproduce. Reporting practice on these benchmarks should change accordingly, and the operator question should be revisited on data with more demonstrators.

REPRODUCIBILITY

The repository, archived on Zenodo at <https://doi.org/10.5281/zenodo.23109850> (all versions), contains the preregistration and amendments with their commit history, the pinned environment, dataset checksums, the immutable partition files, the stored initial conditions, the training and evaluation queue, per-run and per-rollout CSV files for both stages (6,900 and 23,000 rollouts), and the scripts that regenerate every number, table and figure in this paper from those files.

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